



II SEMESTER B. TECH DEGREE ASSIGNMENT
SUBJECT: Engineering Mathematics-II (MAT_1271)

Date: 26/04/2023

Submit by: 05/05/2023

MAX. MARKS: 10

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1. By changing to cylindrical polar coordinates evaluate $\iiint_V z(x^2 + y^2 + z^2) dx dy dz$ where V is the volume of the cylinder $x^2 + y^2 = a^2$ intercepted by the planes $z = 0$ and $z = 4$.
Answer: $4\pi a^2(a^2 + 16)$
 2. Evaluate $\int \int \int \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$ taken throughout the volume of the sphere $x^2 + y^2 + z^2 = 1$ in the positive octant using spherical coordinates.
Answer: $\pi^2/8$
 3. Using double and triple integrals find the volume of the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.
Answer: 16π cc units
 4. Using triple integrals find the volume of the solid bounded by the planes $x = 0, y = 0, x + z = 6$ and $z = 0$.
Answer: 36 cc units
 5. Using triple integrals find the volume of the sphere $x^2 + y^2 + z^2 = a^2$.
Answer: $\frac{4\pi a^3}{3}$ cc units
 6. Using triple integrals find the volume of the paraboloid $x^2 + y^2 = 4z$ cut off by the plane $z = 4$.
Answer: 32π cc units
 7. Using triple integrals find the volume of the solid bounded by the paraboloid $x^2 + y^2 = 2z$ and the cylinder $x^2 + y^2 = 4$.
Answer: 4π cc units
 8. Using triple integrals find the volume of the portion of the sphere $x^2 + y^2 + z^2 = 9$ lying inside the cylinder $x^2 + y^2 = 3y$ above the xy -plane.
Answer: $9\pi - 12$ cc units

9. Evaluate by changing the order of integration $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} dy dx$.

Answer: $\pi a^2/2$

10. Find the volume cut from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$ using triple integration.

Answer: $\frac{2\pi a^3}{3}(2 - \sqrt{2})$

11. Find, by double integration the area lying inside the circle $r = a(\sin \theta)$ and outside the cardioid $r = a(1 - \cos \theta)$.

Answer: $a^2 \left(2 - \frac{\pi}{4}\right)$ square units

12. Evaluate $\iint_R xy\sqrt{1-x-y} dx dy$ where R is the region bounded by

$x = 0, y = 0, x + y = 1$ using $x + y = u, y = uv$.

Answer: $\frac{16}{945}$

13. Using double integrals find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Answer: πab square units

14. Find the area between the curves $\frac{y+8}{x} = x - 2$ and the x -axis.

Answer: 34 square units

15. Find the volume cut from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$.

Answer: $\frac{2\pi a^3}{3}(2 - \sqrt{2})$ cc units

16. Show that $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta \int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \pi^2/2$.

17. Find the Laplace transform of $L[te^{2t} \sin 3t]$.

Answer: $\frac{6(s-2)}{[(s-2)^2+9]^2}$

18. Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 6, & t \geq 2 \end{cases}$ in terms of unit step function hence find the Laplace transform.

Answer: $\frac{2}{s^3} - \frac{2e^{-2s}}{s^3} - \frac{4e^{-2s}}{s^2} + \frac{2e^{-2s}}{s}$

19. Find $L^{-1} \left[\log \left(\frac{s^2+b^2}{s^2+a^2} \right) \right]$.

Answer: $\frac{2(\cos at - \cos bt)}{t}$

20. State and prove convolution theorem.